

Individual Contribution Report

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Master of Applied Computing | University of Windsor

Project: Performance Evaluation of Vector Indexing Techniques for AI Search Systems

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1. Role & Responsibilities

I serve as the Experiments & Visualization Lead for this project. My primary responsibility is to own the benchmarking automation, the experiment configuration system, the scalability experiments, and the visualization that turns raw measurements into clear, comparable results. This role binds the project together: I integrate the data, embedding, indexing, and evaluation components into a single reproducible pipeline and produce the charts and tables that communicate every finding.

Code Modules Owned

- `run_benchmark.py` — command-line entry point and YAML-driven experiment runner (reads the configs/experiment definitions)
- `visualization.py` — comparative and scalability charts plus CSV results-table rendering

Writing Responsibilities

- System Design & Algorithm Overview section of the project proposal
- Implementation Plan and Timeline & Milestones sections of the proposal
- Expected Outcomes and Conclusion sections of the proposal
- Results-visualization and experimental-setup sections of the final report; project README

Technical Responsibilities

- Designing the config-driven benchmarking framework so every experiment is automated and reproducible
- Building the experiment configuration system (configs/) that defines every run
- Designing and running the scalability experiments across corpus sizes from 1K to 100K
- Integrating the full pipeline end-to-end and rendering presentation-grade charts and CSV tables

2. Weekly Contribution Log

The following log records individual actions and outcomes from Week 1 through Week 8, aligned with the course project timeline and phase milestones.

Week 1 — Course Orientation & Background Reading

Actions Taken	- Completed a thorough reading of the course syllabus and grading criteria - Reviewed the Modern Database Architectures module on Brightspace - Conducted independent reading on ANN search and on reproducible benchmarking and experiment automation - Explored Python tooling for automation and visualization (NumPy, Pandas, Matplotlib) to plan the framework
Outcome	Completed Week 1 ready to lead the benchmarking-automation and visualization design.

Week 2 — Team Formation, Role Assignment & Proposal Writing

Actions Taken	- Connected with all three team members — Devu, Ashwin, and Rishi — upon team formation - Participated in the first team meeting to divide responsibilities into four non-overlapping lanes - Took formal ownership of the Experiments & Visualization lane (run_benchmark.py, visualization.py, and the experiment configs) - Set up the shared GitHub repository and the GitHub Projects board for version control and tracking - Began drafting the System Design and Implementation Plan sections of the Pre-Proposal
Outcome	Team roles defined with no overlap; the repository and board were established; system-design writing underway.

Week 3 — Pre-Proposal Submission & Architecture Design

Actions Taken	- Finalized the System Design & Algorithm Overview and Implementation Plan sections of the Pre-Proposal - Produced the proposed benchmarking-framework architecture diagram - Defined the YAML configuration format and command-line interface that would drive every experiment - Participated in full-team review and contributed to the on-time Pre-Proposal submission
Outcome	Pre-Proposal submitted; the system architecture and config-driven experiment plan were finalized.

Week 4 — Peer Review & Benchmarking Tooling Foundation

Actions Taken	• Attended the in-person instructor peer review and presented the framework architecture and automation plan • Incorporated instructor feedback on the System Design into the Full Proposal draft • Implemented the first version of <code>run_benchmark.py</code> — the entry point and YAML-driven experiment runner • Set up the <code>configs/</code> directory and the configuration schema; contributed the System Design section to the Full Proposal
Outcome	Benchmarking entry point and configuration system implemented; System Design sections drafted.

Week 5 — Full Proposal Submitted & Visualization Tooling

Actions Taken	• Finalized the System Design, Implementation Plan, and Timeline sections of the Full Proposal • Implemented <code>visualization.py</code>: comparative bar charts, scalability line charts, and CSV results rendering • Delivered the system-architecture and automation portion in the team's recorded Full-Proposal presentation • Committed the runner, visualization, and configs to the GitHub repository
Outcome	Full Proposal submitted and presented; the runner and visualization tooling were implemented and committed.

Week 6 — Pipeline Integration & Scalability Configuration

Actions Taken	• Integrated the end-to-end pipeline through the config-driven runner (data → embeddings → indexing → query → evaluation → charts) • Built the scalability experiment configurations for corpus sizes from 1K to 100K entries • Ran the first integrated benchmark once the Flat and IVF indexes became available • Coordinated the runner interface with Ashwin (indexing) and Rishi (evaluation); updated the project board
Outcome	End-to-end pipeline integrated and runnable from a single command; scalability configurations prepared.

Week 7 — Initial Result Visualization & Scalability Setup

Actions Taken	- Generated the first comparative and scalability charts from the Flat and IVF benchmark results - Refined the visualization styling (labels, colours, axes) for clear, presentation-ready figures - Built and validated the scalability experiment configurations across the 1K to 100K corpus sizes - Participated in team planning to align the remaining benchmarking timeline
Outcome	Initial result charts produced; the scalability experiment configurations were prepared and validated.

Week 8 — Configuration Finalization & Result Visualization (Current)

Actions Taken	- Finalized the experiment configuration files for the scalability and real-data runs - Produced comparative and scalability charts and CSV result tables to validate the visualization pipeline - Verified that the end-to-end runner integrates cleanly with the latest indexing and evaluation modules - Began drafting the System Design and experimental-setup sections of the final report; updated the project board
Outcome	Benchmarking automation, configuration, and visualization are in place; full scalability runs and final aggregation will follow once HNSW construction completes.

3. Current Project Status

The project is active and ongoing. Phases 1 and 2 are complete and the team is in Phase 3, with HNSW index construction underway and progressing into Phase 4 as initial benchmarking runs for Flat and IVF have begun. My benchmarking-automation, configuration, scalability-experiment, and visualization frameworks are implemented and in place, ready to drive the full set of experiments as the remaining indexes and runs complete.

Phase 1	Dataset Collection & Preprocessing	Complete	ABC News Headlines ingested, cleaned, normalized, and segmented using <code>data_ingestion.py</code> and <code>prepare_headlines.py</code> .
Phase 2	Embedding Generation	Complete	Deterministic 384-dimensional Sentence-BERT embeddings generated via <code>embeddings.py</code> for all corpus sizes.
Phase 3	Index Construction	In Progress	Flat and IVF indexes built and tested; HNSW index currently under construction in FAISS.
Phase 4	Benchmarking & Evaluation	In Progress	Initial benchmarking runs begun for Flat and IVF; HNSW benchmarking to follow as index construction completes.

Phase 5	Analysis & Documentation	Not Started	Report writing, results analysis, and presentation preparation scheduled over the remaining weeks.
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Phase 4 — Benchmarking & Evaluation (My Core Phase)

My automation drives this phase. `run_benchmark.py` reads a YAML configuration and runs the full pipeline; `visualization.py` turns the recorded metrics into comparative and scalability charts and CSV tables. The scalability configurations span corpus sizes from 1K to 100K. Initial Flat and IVF runs and their charts are complete; the full scalability sweeps will run once HNSW construction finishes.

Phase 5 — Analysis & Documentation (Remaining)

Once full results are available, my remaining work is to finalize the result visualizations, write the System Design and experimental-setup sections of the report, complete the README, and assemble the presentation deck.